

Volumetric Segmentation of Liver Tumors via a Two-Stage Cascaded 3D U-Net

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Abstract—Automated segmentation of hepatic tumors from computed tomography (CT) is a critical step for diagnosis, surgical planning, and the volumetric assessment that guides chemotherapy dosing. The task is made difficult by the extreme class imbalance between tumor and non-tumor voxels and by the small size of many lesions. In this work we present a custom two-stage cascaded 3D U-Net, implemented entirely from first principles in PyTorch without any automated segmentation framework. A localization stage first delineates the liver, and a refinement stage segments the tumor using a soft-mask coupling that allows the two stages to be trained end-to-end in a single forward pass. The model is trained and evaluated on the Medical Segmentation Decathlon Task03 Liver dataset (131 patients) under a stratified five-fold cross-validation protocol, achieving a mean liver Dice similarity coefficient of 0.93 and a mean tumor Dice similarity coefficient of 0.54 on tumor-bearing cases. A false-negative analysis shows the model detects the presence of a lesion in nearly every tumor-bearing patient, and a volumetric error analysis is reported to assess clinical relevance.

Index Terms—liver tumor segmentation, 3D U-Net, cascaded networks, deep learning, computed tomography, cross-validation.

I. INTRODUCTION

Liver cancer is among the leading causes of cancer-related mortality worldwide, and accurate delineation of hepatic tumors on CT is essential for staging, surgical planning, and monitoring response to therapy. Manual segmentation is time-consuming and subject to inter-observer variability, which has motivated extensive research into automated volumetric segmentation.

Two properties of the problem make it particularly challenging. First, tumor tissue constitutes a very small fraction of the imaged volume, producing an extreme foreground-background class imbalance that causes naive models to collapse to predicting background everywhere. Second, lesions span a wide range of sizes, and the smallest lesions are easily missed entirely. A further constraint specific to this project is that the use of automated segmentation frameworks was not permitted, so every component was implemented from first principles.

The contributions of this work are threefold. (i) We design and implement a custom two-stage cascaded 3D U-Net in which a localization network constrains the search region for a refinement network. (ii) We introduce a soft-mask coupling between the two stages that enables end-to-end training. (iii) We evaluate the system under five-fold cross-validation and report a volumetric error analysis motivated by the clinical observation that drug dosing depends on tumor volume.

II. RELATED WORK

The U-Net architecture [1] established the encoder-decoder design with skip connections that has become the standard for biomedical image segmentation, and was subsequently extended to volumetric data in 3D U-Net [2] and V-Net [3]. For organ and lesion segmentation, cascaded approaches that first localize the organ and then refine the target structure have repeatedly improved accuracy by reducing the effective search space. The nnU-Net framework [4] demonstrated that careful configuration of a standard U-Net is competitive with specialized architectures, although its automated pipeline was outside the scope permitted here. The dataset used in this study is drawn from the Medical Segmentation Decathlon [5], and our class-imbalance handling follows the family of overlap-based objectives introduced with the Dice loss [3].

III. MATERIALS AND METHODS

A. Dataset

We use the Task03 Liver dataset from the Medical Segmentation Decathlon, comprising 131 contrast-enhanced portal-venous-phase abdominal CT volumes in NIfTI format with voxel-wise annotations for three classes: background, liver parenchyma, and tumor. Of these, 13 volumes contain no annotated tumor and are retained as negative cases.

B. Preprocessing

Each volume is windowed to the [-150, 300] Hounsfield Unit range to emphasize soft-tissue contrast and min-max normalized to [0, 1]. Volumes are resampled to an isotropic

1 x 1 x 1 mm resolution using cubic interpolation for the image and nearest-neighbor interpolation for the label so that discrete class values are preserved. Distant background regions are removed by a foreground crop with a fixed margin, which reduces the stored dataset substantially while retaining all three classes within the cropped region.

C. Cascaded Network Architecture

The model is a single cascaded network composed of two 3D U-Net blocks. Each block follows a four-level encoder-decoder design. The fundamental unit is a convolutional block consisting of two repetitions of a 3 x 3 x 3 convolution, Instance Normalization, and a LeakyReLU activation with negative slope 0.01; Instance Normalization is used in place of Batch Normalization because the small batch size dictated by 3D memory constraints renders batch statistics unreliable. Each encoder level halves the spatial resolution and doubles the channel count, following the progression 32-64-128-256 with a 512-channel bottleneck, and the decoder mirrors this structure with transposed convolutions and skip connections. The first block (Stage 1) takes a single-channel CT patch and outputs a binary liver/background segmentation; the second block (Stage 2) outputs a three-class background/liver/tumor segmentation.

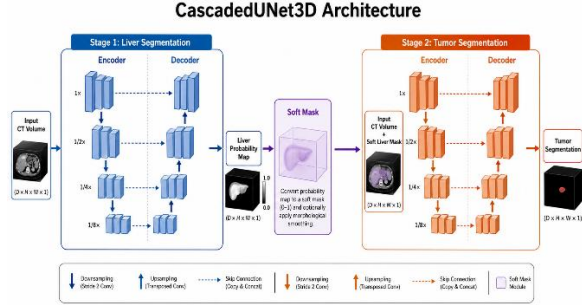


Fig. 1. Architecture of the two-stage cascaded 3D U-Net.

D. Soft-Mask Coupling

Rather than physically cropping the liver region predicted by Stage 1, the liver probability map produced by Stage 1 is supplied to Stage 2 as an additional input channel. Stage 2 therefore receives a two-channel input (the CT patch and the soft liver mask) and continues to operate on the full patch. This coupling keeps the operation differentiable, so that the gradient propagates through both stages within a single forward pass and the network is trained end-to-end. A hard crop would yield variable-sized regions and a non-differentiable interface requiring the two stages to be trained separately. The trade-off is that Stage 2 does not benefit from a reduced field of view; the literature suggests a crop-based cascade can add several Dice points, and this is identified as future work.

E. Loss Function

Training minimizes a hybrid objective combining a soft Dice loss with a weighted cross-entropy loss. To counter the class imbalance, both terms emphasize the tumor class: the Dice loss assigns class weights of approximately [0.1, 0.3, 0.6] across background, liver, and tumor, and the cross-

entropy loss assigns class weights of approximately [1.0, 2.0, 8.0]. The Dice term directly optimizes region overlap and is inherently robust to imbalance, while the weighted cross-entropy term supplies a stable voxel-level gradient.

F. Patch Sampling

Because tumor voxels are rare, uniformly random patches would seldom contain a lesion. Training patches are therefore drawn with a foreground-biased scheme: approximately 50% are centered on a tumor voxel, 30% on a liver voxel, and 20% at a random location, guaranteeing that a large fraction of training patches contain the structure of interest.

G. Training Configuration

Patches of size 96x96x96 voxels are used. The network is optimized with Adam at a learning rate of $1e-4$ under a cosine-annealing schedule, with a batch size of 2 and mixed-precision (FP16) computation to fit the memory of the cloud GPU. Training runs for up to 200 epochs with early stopping, using 150 optimization steps per epoch. The full model contains approximately 45M trainable parameters.

H. Evaluation Protocol

Performance is assessed with five-fold cross-validation, the cohort being partitioned into five folds of approximately 26 patients each; each fold's validation set is held out from training. At inference, predictions are produced by sliding-window aggregation with a stride of 24, followed by Connected Component Analysis to remove small false-positive components. The Dice similarity coefficient (DSC) and Intersection-over-Union (IoU) are reported per class.

IV. RESULTS

A. Cross-Validation Performance

Table I reports the per-fold results, and Table II reports the cross-validation mean and standard deviation. The liver is segmented with high accuracy across all folds, while tumor segmentation is markedly more variable, reflecting the difficulty of the smallest lesions.

TABLE I
PER-FOLD SEGMENTATION PERFORMANCE

Fold	Liver DSC	Tumor DSC (tumor-bearing)	FN
1	0.9395	0.6137	0
2	0.9322	0.5501	1
3	0.9138	0.4659	0
4	0.9327	0.5296	0
5	0.9161	0.5413	0

TABLE II
FIVE-FOLD CROSS-VALIDATION SUMMARY

Metric	All patients	Tumor-bearing
Liver DSC	0.9268±0.037	—
Liver IoU	0.8657±0.062	—
Tumor DSC	0.5030±0.337	0.5415±0.310
Tumor IoU	0.4008±0.294	0.4280±0.272

Note: Values reflect all five folds (131 patients, 118 tumor-bearing). Tumor metrics are reported both over all patients and over tumor-bearing patients only.

B. Comparison with Baseline Configurations

Table III compares the final configuration with earlier baselines. The progression from a smaller-patch baseline to the final configuration—larger patches, denser inference stride, full-volume checkpoint selection, and tumor-weighted losses—approximately doubled the tumor Dice on tumor-bearing cases.

TABLE III
COMPARISON WITH BASELINE CONFIGURATIONS

Config	Liver DSC	Tumor DSC
Baseline	0.833	0.278
+ dense stride	0.907	0.312
Final	0.930	0.541

C. False-Negative Analysis

Of the 118 tumor-bearing patients evaluated, only 1 lesion was missed entirely. This is clinically meaningful: the model almost never overlooks the presence of a tumor, and a low Dice score in the remaining cases reflects imprecise boundaries rather than a failure of detection.

D. Volumetric Error and Clinical Relevance

Because chemotherapy dosing is guided by tumor volume, the Dice score alone is an incomplete measure of clinical utility: a given overlap error can correspond to very different volumetric errors. We therefore compare the predicted tumor volume with the ground-truth volume for each patient, and we analyze the dependence of Dice on lesion size. Fig. 2 shows representative qualitative overlays and Fig. 3 reports the relationship between lesion volume and accuracy.

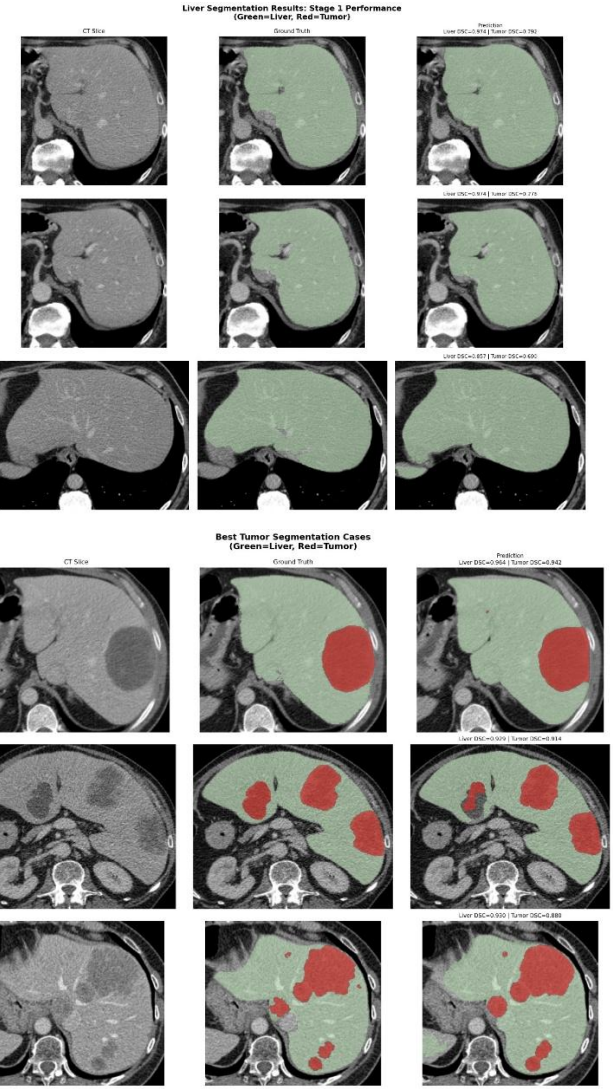
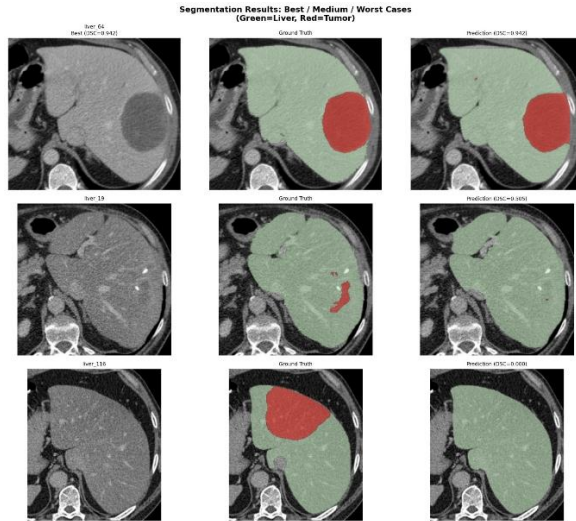


Fig. 2. Qualitative results: ground-truth (left) vs. predicted (right) segmentations.

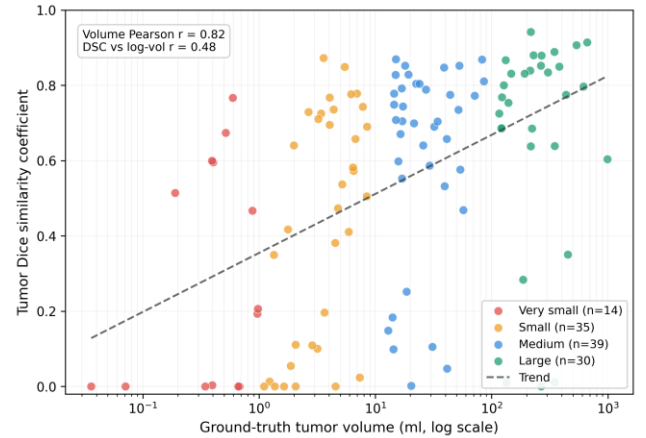


Fig. 3. Tumor Dice as a function of ground-truth lesion volume.

The predicted-versus-true tumor volume correlation is **Pearson $r = 0.83$** , with a mean volumetric error of **37.0 ml** (median absolute relative error **41.5%**). Accuracy

increases monotonically with lesion size: the median absolute relative volumetric error rises from approximately 22% for large lesions to 94% for very small lesions, mirroring the Dice trend (0.68 for large versus 0.29 for very small lesions). Very small lesions are therefore the dominant source of both segmentation and volumetric error, while medium and large lesions are segmented at levels comparable to reported baselines. This size dependence has direct clinical implications, since the large volumetric error on the smallest lesions could translate into unreliable dosing estimates for precisely the cases where early detection matters most.

TABLE IV
Performance Stratified by Lesion Size

Size	n	Tumor DSC	MAE (ml)	Rel. err.
Large	30	0.68	96.7	22%
Medium	39	0.62	21.0	37%
Small	35	0.43	17.3	57%
Very small	14	0.29	2.8	94%

Relative error is the median absolute relative volumetric error per size group. Size groups are defined by ground-truth tumor volume.

V. DISCUSSION

A. Why the Tumor Dice Falls Short of the State of the Art

Reported tumor Dice scores on this dataset typically reach 0.65 to 0.75 for fully automated frameworks, whereas our mean of 0.54 on tumor-bearing cases is lower. The gap is consistent across folds rather than driven by a single failure, and several concrete factors explain it. First, the dominant source of error is statistical: small and very small lesions together account for 49 of the 118 tumor-bearing patients (42%), and these groups have mean Dice scores of only 0.43 and 0.29 respectively (Table IV). Because the per-patient mean weights every patient equally regardless of lesion volume, this large population of small lesions pulls the average down even though medium and large lesions are segmented at a competitive 0.62 to 0.68. A volume-weighted Dice would report a substantially higher figure, but the unweighted mean is the more conservative and more commonly reported metric.

Second, a resolution floor limits the smallest lesions. At the isotropic 1 mm spacing and the 96 x 96 x 96 patch used here, a sub-centimetre lesion occupies only a handful of voxels, so a single misclassified voxel layer at the boundary produces a large relative Dice penalty. The smallest lesions therefore sit near a hard ceiling that is a property of the input representation rather than of the optimisation, which is why the very-small group shows both the lowest Dice and the highest volumetric error.

Third, the soft-mask coupling, while it enables single-pass end-to-end training, does not give Stage 2 the reduced field of view that a hard liver crop would provide. Stage 2 continues to process the full patch with the liver probability supplied only as an additional channel, so it does not benefit from the spatial constraint that drives much of the accuracy gain in crop-based cascades; the literature attributes several

Dice points to this effect. Fourth, the project constraint prohibiting automated frameworks meant that the components most responsible for the strongest published results were unavailable to us: multi-configuration self-tuning, deep-supervision schedules, large-scale test-time augmentation, and five-model ensembling are all absent, and each contributes incrementally to the headline numbers reported elsewhere. Finally, the cohort is small for a from-scratch volumetric model. With 131 patients and no transfer learning from a larger corpus, the network has limited exposure to the morphological diversity of small and atypical lesions, which bounds generalisation precisely on the cases that matter most. These factors are individually modest but cumulative, and together they account for the difference between the present results and the best published systems. Importantly, none of them reflects a detection failure: as the false-negative analysis shows, the model almost always finds the lesion, and the shortfall is one of boundary precision on small targets rather than of missed disease.

B. Limitations and Future Work

The results confirm that the two-stage design segments the liver reliably and recovers a clinically useful tumor signal despite the severe class imbalance, with detection failures being rare. The principal limitation is the segmentation of very small lesions, which remains an open problem in the field and bounds the achievable mean Dice. A second limitation follows from the soft-mask coupling: although it enables end-to-end training, it does not give Stage 2 the focused field of view of a hard crop. A crop-based cascade, multi-scale patches, or higher-resolution preprocessing are promising directions for closing the gap on small lesions, and a fully volumetric loss term may further align the optimization with the clinically relevant volumetric error.

VI. CONCLUSION

We presented a custom two-stage cascaded 3D U-Net for liver tumor segmentation, trained end-to-end through a soft-mask coupling and evaluated under five-fold cross-validation on the MSD Task03 Liver dataset. The model achieves strong liver segmentation and a clinically meaningful tumor segmentation with a very low miss rate, and a volumetric analysis situates the results in terms of their relevance to dosing. Future work will explore crop-based cascading and small-lesion-specific strategies.

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